

Chapter 4 Sampling distributions

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4.1 Introduction

- Sample

A set of observable random variables $X_1, \dots, X_n$ with sample size n

- Random sample

Independent and identically distributed (iid) observable random variables $X_1, \dots, X_n$

- Statistic T

A function T of observable random variables $X_1, \dots, X_n$ that does not depend on any unknown parameters

- Sampling distribution

The probability distribution of a sample statistic T

4.1 Introduction

- **Sampling distribution**: the probability distribution of a sample statistic
 - sample mean
 - sample variance
- Sampling distributions provide the **link** between probability theory and statistical inference
- In general, a population has a distribution called a **population distribution**, which is usually **unknown**
- A statistic has a sampling distribution, which is usually different from the population probability distribution
- The sampling distribution of a statistic provides a theoretical model of the relative frequency histogram for the likely values of the statistic that one would observe through repeated sampling

4.1 Introduction

Example 4.1.1 Let the population consist of the numbers 1, 2, 3, 4, 5. Consider all possible samples consisting of three numbers randomly chosen without replacement from this population. Obtain the distribution of the sample mean.

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Solution

Disregarding the order, it is clear that there are $\binom{5}{3} = 10$ equally likely possible samples of size 3. They are (1, 2, 3), (1, 2, 4), (1, 2, 5), (1, 3, 4), (1, 3, 5), (1, 4, 5), (2, 3, 4), (2, 3, 5), (2, 4, 5), and (3, 4, 5). Calculating the sample mean, $\bar{X}$, for each of the samples, we will get the sampling distribution of $\bar{X}$ as:

$\bar{x}$	2/1	7/3	8/3	3/1	10/3	11/3	4/1
$p(\bar{x})$	$\frac{1}{10}$	$\frac{1}{10}$	$\frac{2}{10}$	$\frac{2}{10}$	$\frac{2}{10}$	$\frac{1}{10}$	$\frac{1}{10}$

4.1 Introduction

Theorem 4.1.1 Let $X_1, \dots, X_n$ be a random sample of size n from a population with mean μ and variance σ^2 . Then $E(\bar{X}) = \mu$ and $Var(\bar{X}) = \frac{\sigma^2}{n}$.

- The sample means become more and more reliable as an estimate of μ as the sample size is increased, as we would expect

$$P \{ |\bar{X} - \mu| < K\sigma_{\bar{X}} \} \geq 1 - \frac{1}{K^2}$$

$$P \{ |\bar{X} - \mu| < \epsilon \} \geq 1 - \frac{\sigma^2}{n\epsilon^2}$$

- For any $\epsilon > 0$, the probability that the difference between $\bar{X}$ and μ is less than ϵ can be made arbitrarily close to 1 by choosing sample size n that is sufficiently large.

4.1 Introduction

Example 4.1.2 A particular brand of drink is packaged at an average of 12 oz per bottle. As a result of randomness, there will be small variations in how much liquid each bottle really contains. It has been observed that the amount of liquid in these bottles is normally distributed with $\sigma = 0.8$ oz. A sample of 10 bottles of this brand of soda is randomly selected from a large lot of bottles, and the amount of liquid, in ounces, is measured in each. Find the probability that the sample mean will be within 0.5 oz of 12 oz.

4.1 Introduction

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Solution

Let $X_1, X_2, \dots, X_{10}$ denote the ounces of liquid measured for each of the bottles. We know that X_i 's are normally distributed with mean $\mu = 12$ and variance $\sigma^2 = 0.64$. From [Theorem 4.1.1](#), $\bar{X}$ possesses a normal distribution (actually, for the normality part, we use [Corollary 4.2.2](#)) with a mean 12 and variance $\sigma^2/n = 0.64/10 = 0.064$.

4.1 Introduction

$$\begin{aligned}P(|\bar{X} - 12| \leq 0.5) &= P(-0.5 \leq \bar{X} - 12 \leq 0.5) \\&= P\left(-\frac{0.5}{\sigma/\sqrt{n}} \leq \frac{\bar{X} - 12}{\sigma/\sqrt{n}} \leq \frac{0.5}{\sigma/\sqrt{n}}\right) \\&= P\left(-\frac{0.5}{0.253} \leq Z \leq \frac{0.5}{0.253}\right) \\&= P(-1.97 \leq Z \leq 1.97) = 0.9512\end{aligned}$$

Hence, the chance is about 0.95% that the mean amount of drink in any 10 bottles randomly chosen will be between 11.5 and 12.5 oz.

4.1 Introduction

Example 4.1.4 Let $X_1, \dots, X_n$ be a random sample of size n from a population with mean μ and variance σ^2 . Consider the sample variance:

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

Show that $E(S^2) = \sigma^2$.

4.1 Introduction

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Show that $E(S^2) = \sigma^2$.

Solution

$$\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 = \frac{\sum_{i=1}^n X_i^2 - n\bar{X}^2}{n-1}.$$

$$E(S^2) = E\left(\frac{\sum_{i=1}^n X_i^2 - n\bar{X}^2}{n-1}\right)$$

$$= \frac{1}{n-1} \sum_{i=1}^n E(X_i^2) - \frac{n}{n-1} E(\bar{X}^2)$$

4.1 Introduction

Using the fact that $E(X^2) = \text{Var}(X) + \mu^2$ and [Theorem 4.1.1](#), we have:

$$\begin{aligned} E(S^2) &= \frac{n}{n-1} (\sigma^2 + \mu^2) - \frac{n}{n-1} \left(\frac{\sigma^2}{n} + \mu^2 \right) \\ &= \left(\frac{n}{n-1} - \frac{1}{n-1} \right) \sigma^2 - \left(\frac{n}{n-1} - \frac{n}{n-1} \right) \mu^2 = \sigma^2. \end{aligned}$$

- The expected value of the sample variance is the same as the variance of the population under consideration

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4.2 Sampling distributions for normal populations

Theorem 4.2.1 Let $X_1, \dots, X_n$ be independent random variables with the distribution of X_i being normal with mean μ_i and variance σ_i^2 . Let $a_1, a_2, \dots, a_n$ be real constants. Then the distribution of

$$Y = \sum_{i=1}^n a_i X_i$$

is normal with mean

$$\mu_Y = \sum_{i=1}^n a_i \mu_i$$

and variance

$$\sigma_Y^2 = \sum_{i=1}^n a_i^2 \sigma_i^2$$

- **Proof.** moment-generating (mgf) function of Y

4.2 Sampling distributions for normal populations

Corollary 4.2.2 Let $X_1, \dots, X_n$ be a random sample of size n from a normal population with mean μ and variance σ^2 . Then:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

is normally distributed with mean

$$\mu_{\bar{X}} = \mu$$

and variance

$$\sigma_{\bar{X}}^2 = \frac{\sigma^2}{n}$$

4.2 Sampling distributions for normal populations

Example 4.2.1 A company that manufactures cars claims that the gas mileage for its new line of hybrid cars, on the average, is 60 miles per gallon with a standard deviation of 4 miles per gallon. A random sample of 16 cars yielded a mean of 57 miles per gallon. If the company's claim is correct, what is the probability that the sample mean is less than or equal to 57 miles per gallon? Comment on the company's claim about the mean gas mileage per gallon of its cars. What assumptions did you make?

4.2 Sampling distributions for normal populations

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Solution

Let X represent the gas mileage for the new car (in miles per gallon). If the company's claim is true, then from Corollary 4.2.2, X is normally distributed with mean $\mu = 60$ and variance $\sigma^2/n = 16/16 = 1$.

4.2 Sampling distributions for normal populations

Hence,

$$\begin{aligned} P(\bar{X} \leq 57) &= P\left(\frac{\bar{X} - 60}{1} \leq \frac{57 - 60}{1}\right) \\ &= P(Z \leq -3) \approx 1 - 0.999 = 0.001. \end{aligned}$$

- If the company's claim is correct, it is very unlikely that the mean value of the random sample of 16 cars will be 57 miles per gallon
- Because the mean is indeed 57 miles per gallon, we conclude that the company's claim is very likely not true
- Here we have assumed that the sample of 16 measurements comes from a normal population, so that we could apply the results of Corollary 4.2.2

4.2.1. Chi-square distribution

- The chi-square distribution is a special case of a gamma distribution with $\alpha = n/2$ and $\beta = 2$
- If n is a positive integer, it is called the degrees of freedom
- If n is not an integer, but $\beta = 2$, we still refer to this distribution as a chi-square.
- The mgf of a χ^2 -random variable with n degrees of freedom is

$$M(t) = (1 - 2t)^{-n/2}$$

- If $X \sim \chi^2(n)$, then $E(X) = n$ and $Var(X) = 2n$
- Let $X_1, \dots, X_n$ be independent χ^2 -random variables with $n_1, \dots, n_k$ degrees of freedom, respectively. Let $V = \sum_{i=1}^k X_i$, then

$$V \sim \chi^2 \left(\sum_{i=1}^k n_i \right)$$

4.2.1. Chi-square distribution

- **Theorem 4.2.4** Let X_1 and X_2 be independent random variables. Suppose that $X_1 \sim \chi^2(n_1)$, whereas $Y = X_1 + X_2 \sim \chi^2(n)$, where $n > n_1$. Then $X_2 = Y - X_1 \sim \chi^2(n - n_1)$.
- **Theorem 4.2.5** If a random variable X has a gamma probability distribution with parameters α and β , then:

$$Y = \frac{2X}{\beta} \sim \chi^2(2\alpha)$$

- **Theorem 4.2.6** If X is a standard normal random variable, then X^2 is chi-square random variable with 1 degree of freedom.

4.2.1. Chi-square distribution

- **Theorem 4.2.7** Let the random sample $X_1, \dots, X_n$ be from an $N(\mu, \sigma^2)$ distribution. Then $Z_i = \frac{X_i - \mu}{\sigma}$ are independent standard normal random variables and

$$\sum_{i=1}^n Z_i^2 = \sum_{i=1}^n \left(\frac{X_i - \mu}{\sigma} \right)^2 \sim \chi^2(n)$$

- **Upper quantile ($\chi_\alpha^2(n)$ or $\chi_{\alpha,n}^2$)**

If $X \sim \chi^2(n)$, the values of $\chi_\alpha^2(n)$ such that:

$$P(X > \chi_\alpha^2(n)) = \alpha$$

4.2.1. Chi-square distribution

Example 4.2.2 Let the random variables $X_1, X_2, \dots, X_5$ be from an $N(5, 1)$ distribution. Find a number a such that

$$P\left(\sum_{i=1}^5 (X_i - 5)^2 \leq a\right) = 0.90 :$$

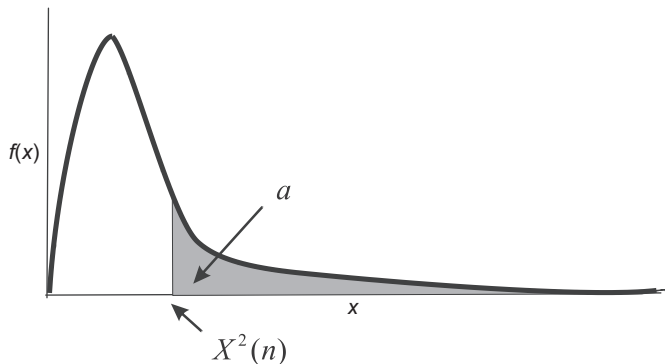


FIGURE 4.2 Chi-square probability density.

4.2.1. Chi-square distribution

Solution

$$\sum_{i=1}^5 Z_i^2 = \sum_{i=1}^5 \frac{(X_i - 5)^2}{1^2} = \sum_{i=1}^5 (X_i - 5)^2 \sim \chi^2(5)$$

Because the upper tail area is 0.10, looking at the chi-square table with 5 degrees of freedom and the column corresponding to $\chi_{0.10}^2$, we obtain $a = 9.23635$. Thus,

$$P\left(\sum_{i=1}^5 (X_i - 5)^2 \leq 9.23635\right) = 0.90.$$

4.2.1. Chi-square distribution

Example 4.2.3 Suppose that X is a chi-square random variable with 20 degrees of freedom. Use the chi-square table to obtain the following:

- (a) Find x_0 such that $P(X > x_0) = 0.95$.
- (b) Find $P(X \leq 12.443)$.

4.2.1. Chi-square distribution

Example 4.2.3 Suppose that X is a chi-square random variable with 20 degrees of freedom. Use the chi-square table to obtain the following:

- (a) Find x_0 such that $P(X > x_0) = 0.95$.
- (b) Find $P(X \leq 12.443)$.

Solution

- (a) For 20 degrees of freedom, using the chi-square table, we have:

$$P(X > 10.851) = 0.95.$$

Hence, $x_0 = 10.851$.

- (b) From the chi-square table,

$$P(X \leq 12.443) = 0.10.$$

4.2.1. Chi-square distribution

Theorem 4.2.8 If $X_1, \dots, X_n$ is a random sample from a normal population with the mean μ and variance σ^2 , then:



$$\frac{\sum_{i=1}^n (X_i - \bar{X})^2}{\sigma^2} = \frac{(n-1)S^2}{\sigma^2} \sim \chi^2(n-1)$$

- $\bar{X}$ and S^2 are independent

4.2.1. Chi-square distribution

Example 4.2.4 Let $X_1, X_2, \dots, X_{10}$ be a random sample from a normal distribution with $\sigma^2 = 0.8$. Find two positive numbers a and b such that the sample variance S^2 satisfies

$$P(a \leq S^2 \leq b) = 0.90.$$

4.2.1. Chi-square distribution

Example 4.2.4 Let $X_1, X_2, \dots, X_{10}$ be a random sample from a normal distribution with $\sigma^2 = 0.8$. Find two positive numbers a and b such that the sample variance S^2 satisfies

$$P(a \leq S^2 \leq b) = 0.90.$$

Solution

Because $(n-1)S^2/\sigma^2 \sim \chi^2(n-1)$, we have

$$P(a \leq S^2 \leq b) = P\left(\frac{(n-1)a}{\sigma^2} \leq \frac{(n-1)S^2}{\sigma^2} \leq \frac{(n-1)b}{\sigma^2}\right).$$

The desired values can be found by setting the upper tail area and lower tail area each equal to 0.05. Using the chi-square table with $n-1 = 9$ degrees of freedom, we have:

$$\frac{(n-1)b}{\sigma^2} = \frac{9b}{0.8} = \chi_{0.05}^2(9) = 16.919,$$

4.2.1. Chi-square distribution

which implies $b = \frac{(16.919) \times (0.8)}{9} = 1.50$. Similarly,

$$\frac{(n-1)a}{\sigma^2} = \frac{9a}{0.8} = \chi_{0.95}^2(9) = 3.325.$$

So we have $a = \frac{(3.325) \times (0.8)}{9} = 0.295$.

Hence,

$$P(0.295 \leq S^2 \leq 1.50) = 0.90.$$

- This is **not the only** interval that would satisfy $P(a \leq S^2 \leq b) = 0.90$, but it is a convenient one.

4.2.1. Chi-square distribution

Example 4.2.5 A fruit-drink company wants to know the variation, as measured by the standard deviation, of the amount of juice in 16-oz cans. From past experience, it is known that $\sigma^2 = 2$. The company statistician decides to take a sample of 25 cans from the production line and compute the sample variance. Assuming that the sample values may be viewed as a random sample from a normal population, find a value of b such that $P(S^2 > b) = 0.05$.

4.2.1. Chi-square distribution

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Solution

$$(n-1)S^2/\sigma^2 \sim \chi^2(n-1) = \chi^2(24)$$

$$0.05 = P(S^2 > b) = P\left(\frac{24S^2}{2} > \frac{24b}{2}\right) = P(\chi^2 > c),$$

$$c = 36.4151, b = \frac{2}{24}, c = \frac{2}{24}(36.4151) = 3.03$$

$$P(S^2 > 3.03) = 0.05$$

4.2.2 Student t distribution

Definition 4.2.2 If Y and Z are independent random variables, Y has a chi-square distribution with n degrees of freedom, and $Z \sim N(0, 1)$, then:

$$T = \frac{Z}{\sqrt{\frac{Y}{n}}}$$

is said to have a (Student) t -distribution with n degrees of freedom. We denote this by $T \sim T_n$.

The probability density of the random variable T with n degrees of freedom is given by:

$$f(t) = \frac{\Gamma(\frac{n+1}{2})}{\sqrt{\pi n} \Gamma(\frac{n}{2})} \left(1 + \frac{t^2}{n}\right)^{-\frac{n+1}{2}}, -\infty < t < \infty$$

4.2.2 Student t distribution

- If $T \sim T_n$, $E(T) = 0$ and $Var(T) = \frac{n}{n-2}, n > 2$
- If $T \sim T_n$, the value of $t_{\alpha,n}$ is such that $P(t > t_{\alpha,n}) = \alpha$

Theorem 4.2.9 If $\bar{X}$ and S^2 are the mean and the variance of a random sample of size n from a normal population with mean μ and variance σ^2 , then:

$$T = \frac{\bar{X} - \mu}{\frac{S}{\sqrt{n}}}$$

has a t distribution with $n - 1$ degrees of freedom.

4.2.2 Student t distribution

Example 4.2.6 A manufacturer of fuses claims that with a 20% overload, the fuses will blow in less than 10 minutes on average. To test this claim, a random sample of 20 of these fuses was subjected to a 20% overload, and the times it took them to blow had a mean of 10.4 minutes and a sample standard deviation of 1.6 minutes. It can be assumed that the data constitute a random sample from a normal population. Do they tend to support or refute the manufacturer's claim?

4.2.2 Student t distribution

Example 4.2.6 A manufacturer of fuses claims that with a 20% overload, the fuses will blow in less than 10 minutes on average. To test this claim, a random sample of 20 of these fuses was subjected to a 20% overload, and the times it took them to blow had a mean of 10.4 minutes and a sample standard deviation of 1.6 minutes. It can be assumed that the data constitute a random sample from a normal population. Do they tend to support or refute the manufacturer's claim?

Solution

Given $\bar{y} = 10.4$, $s = 1.6$, $n = 20$, and $\mu = 10$, hence,

$$t = \frac{\bar{y} - \mu}{s/\sqrt{n}} = \frac{10.4 - 10}{1.6/\sqrt{20}} = 1.118.$$

The degrees of freedom is $n - 1 = 19$. From the t -table, the probability that t exceeds 1.328 is 0.10. Since the observed value of $t = 1.118$ is less than $t_{0.10,19} = 1.328$, and 0.10 is a relatively large probability, we conclude that the data tend to support the manufacturer's claim.

4.2.2 Student t distribution

Example 4.2.7 The human gestation period - the period of time between conception and labor - is approximately 40 weeks (280 days), measured from the first day of the mother's last menstrual period. For a newborn full-term infant, the appropriate length for gestational age is assumed to be normally distributed with $\mu = 50\text{cm}$ and $\sigma = 1.25\text{cm}$. Compute the probability that a random sample of 20 infants born at full term results in a sample mean greater than 52.5cm.

4.2.2 Student t distribution

Example 4.2.7 The human gestation period - the period of time between conception and labor - is approximately 40 weeks (280 days), measured from the first day of the mother's last menstrual period. For a newborn full-term infant, the appropriate length for gestational age is assumed to be normally distributed with $\mu = 50\text{cm}$ and $\sigma = 1.25\text{cm}$. Compute the probability that a random sample of 20 infants born at full term results in a sample mean greater than 52.5cm.

Solution

- $\bar{X} \sim N(50, 1.56/20)$
- Even though the sample size is small, since σ is known, we do not use the t -distribution
- $P(\bar{X} > 52.5) = P\left(z > \frac{52.5-50}{1.25/\sqrt{20}} = 8.94\right) \approx 0$
- $P(X > 52.5) \approx 0.0228$

4.2.3 F -distribution

Definition 4.2.3 Let U and V be chi-square random variables with n_1 and n_2 degrees of freedom, respectively. Then if U and V are independent,

$$F = \frac{\frac{U}{n_1}}{\frac{V}{n_2}}$$

is said to have an F -distribution with n_1 numerator degrees of freedom and n_2 denominator degrees of freedom. We denote this by $F \sim F(n_1, n_2)$

The pdf for a random variable $F \sim F(n_1, n_2)$ is given by:

$$f(x) = \begin{cases} \frac{\Gamma\left(\frac{n_1+n_2}{2}\right)}{\Gamma\left(\frac{n_1}{2}\right)\Gamma\left(\frac{n_2}{2}\right)} \left(\frac{n_1}{n_2}\right)^{\frac{n_1}{2}} x^{\frac{n_1}{2}-1} \left(1 + \frac{n_1}{n_2}x\right)^{-\frac{(n_1+n_2)}{2}}, & x > 0 \\ 0, & \text{elsewhere} \end{cases}$$

$$P(F > F_\alpha(n_1, n_2)) = \alpha, \text{ and } F_{1-\alpha}(n_2, n_1) = \frac{1}{F_\alpha(n_1, n_2)}$$

4.2.3 F -distribution

Theorem 4.2.10 Let two independent random samples of size n_1 and n_2 be drawn from two normal populations with variances σ_1^2, σ_2^2 respectively. If the variances of the random samples are given by S_1^2, S_2^2 , respectively, then the statistic:

$$F = \frac{\frac{S_1^2}{\sigma_1^2}}{\frac{S_2^2}{\sigma_2^2}} = \frac{\sigma_2^2 S_1^2}{\sigma_1^2 S_2^2} \sim F(n_1 - 1, n_2 - 1)$$

Corollary 4.2.11 If $\sigma_1^2 = \sigma_2^2$, then

$$F = \frac{S_1^2}{S_2^2} \sim F(n_1 - 1, n_2 - 1)$$

- When $\sigma_1^2 = \sigma_2^2$, we refer to them as two populations that are homogeneous with respect to their variances.

4.2.3 F -distribution

Example 4.2.8 Let S_1^2 denote the sample variance for a random sample of size 10 from population I, and let S_2^2 denote the sample variance for a random sample of size 8 from population II. The variance of population I is assumed to be three times the variance of population II. Find two numbers a and b such that $P(a \leq S_1^2/S_2^2 \leq b) = 0.90$ assuming S_1^2 to be independent of S_2^2 .

4.2.3 F -distribution

Example 4.2.8 Let S_1^2 denote the sample variance for a random sample of size 10 from population I, and let S_2^2 denote the sample variance for a random sample of size 8 from population II. The variance of population I is assumed to be three times the variance of population II. Find two numbers a and b such that $P(a \leq S_1^2/S_2^2 \leq b) = 0.90$ assuming S_1^2 to be independent of S_2^2 .

Solution

From the problem, we can assume that $\sigma_1^2 = 3\sigma_2^2$ with $n_1 = 10$ and $n_2 = 8$. Thus, we can write:

$$\frac{\frac{S_1^2}{\sigma_1^2}}{\frac{S_2^2}{\sigma_2^2}} = \frac{\frac{S_1^2}{3\sigma_2^2}}{\frac{S_2^2}{\sigma_2^2}} = \frac{S_1^2}{3S_2^2}$$

which has an F -distribution with $n_1 - 1 = 9$ numerator and $n_2 - 1 = 7$ denominator degrees of freedom.

4.2.3 F -distribution

Using the F -table, $F_{0.05}(9, 7) = 3.68$. Now to find $F_{0.95}(9, 7)$ such that:

$$P\left(\frac{S_1^2}{3S_2^2} < F_{0.95}(9, 7)\right) = 0.05$$

We proceed as follows:

$$P\left(\frac{S_1^2}{3S_2^2} < F_{0.95}(9, 7)\right) = P\left(\frac{3S_2^2}{S_1^2} > \frac{1}{F_{0.95}(9, 7)}\right) = 0.05$$

Indexing $v_1 = 7$ and $v_2 = 9$ in the F -table, we have $1/F_{0.95}(9, 7) = F_{0.05}(7, 9) = 3.29$. We get $F_{0.95}(9, 7) = 0.304$. Hence, the entire probability statement is given by:

$$P\left(0.304 \leq \frac{S_1^2}{3S_2^2} \leq 3.68\right) = P\left(0.912 \leq \frac{S_1^2}{S_2^2} \leq 11.04\right) = 0.90$$

Thus, $a = 0.912$ and $b = 11.04$.

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4.3 Order statistics

Definition 4.3.1 Let $X_1, \dots, X_n$ be a random sample from a continuous distribution with pdf $f(x)$. Let $Y_1, \dots, Y_n$ be a permutation of $X_1, \dots, X_n$ such that

$$Y_1 \leq Y_2 \leq \dots \leq Y_n$$

Then the ordered random variables $Y_1, \dots, Y_n$ are called the order statistics of the random sample $X_1, \dots, X_n$. Here Y_k is called the k th order statistic.

- The order statistics of the sample $X_1, X_2, \dots, X_n$ can also be denoted by $X_{(1)}, X_{(2)}, \dots, X_{(n)}$ where

$$X_{(1)} < X_{(2)} < \dots < X_{(n)}$$

- The minimum of X_i

$$Y_1 = \min(X_1, \dots, X_n)$$

- The maximum is

$$Y_n = \max(X_1, \dots, X_n)$$

4.3 Order statistics

Example 4.3.1

- The range $R = Y_n - Y_1$ is a function of order statistics.
- The sample median M equals Y_{m+1} if $n = 2m + 1$.
 - When n is odd, the sample median M is an order statistic
 - When n is even, then the sample median can be obtained using the order statistic

$$M = \frac{1}{2} \left(Y_{\frac{n}{2}} + Y_{(\frac{n}{2})+1} \right).$$

4.3 Order statistics

Theorem 4.3.1 Let $X_1, \dots, X_n$ be a random sample from a population with pdf $f(x)$. Then the joint pdf of order statistics $Y_1, \dots, Y_n$ is:

$$f(y_1, \dots, y_n) = \begin{cases} n! f(y_1) f(y_2) \dots f(y_n), & \text{for } y_1 < \dots < y_n \\ 0, & \text{otherwise} \end{cases}$$

Theorem 4.3.2 The pdf of Y_k is:

$$f_k(y) = f_{Y_k} = \frac{n!}{(k-1)!(n-k)!} f(y) (F(y))^{k-1} (1 - F(y))^{n-k}$$

for $-\infty < y < \infty$, where $F(y) = P(X_i \leq y)$ is the cumulative distribution function (cdf) of X_i .

4.3 Order statistics

Example 4.3.2 Let $X_1, \dots, X_n$ be a random sample from $U[0, 1]$. Find the pdf of the k_{th} order statistic Y_k .

4.3 Order statistics

Example 4.3.2 Let $X_1, \dots, X_n$ be a random sample from $U[0, 1]$. Find the pdf of the k_{th} order statistic Y_k .

Solution

Since the pdf of X_i is $f(x) = 1, 0 \leq x \leq 1$, the cdf is $F(x) = x, 0 \leq x \leq 1$. Using [Theorem 4.3.2](#), the pdf of the k_{th} order statistic Y_k reduces to:

$$f_k(y) = \frac{n!}{(k-1)!(n-k)!} y^{k-1} (1-y)^{n-k}, \quad 0 \leq y \leq 1$$

which is a beta distribution with $\alpha = k$ and $\beta = n - k + 1$.

4.3 Order statistics

Example 4.3.3 Find the distribution of the n th order statistic Y_n of the sample $X_1, \dots, X_n$ from a population with pdf $f(x)$.

4.3 Order statistics

Example 4.3.3 Find the distribution of the n th order statistic Y_n of the sample $X_1, \dots, X_n$ from a population with pdf $f(x)$.

Solution

Let the cdf of Y_n be denoted by $F_n(y)$. Then:

$$\begin{aligned} F_n(y) &= P(Y_n \leq y) = P\left(\max_{1 \leq i \leq n} X_i \leq y\right) \\ &= P(X_1 \leq y, \dots, X_n \leq y) = [F(y)]^n \quad (\text{by independence}). \end{aligned}$$

$$f_n(y) = \frac{d}{dy} [F(y)^n] = n[F(y)]^{n-1} \frac{d}{dy} F(y) = n[F(y)]^{n-1} f(y).$$

In particular, if $X_1, \dots, X_n \sim U[0, 1]$, then:

$$F_n(y) = \begin{cases} 0, & \text{if } y < 0 \\ y^n, & \text{if } 0 \leq y \leq 1 \\ 1, & \text{if } y > 1 \end{cases}$$

4.3 Order statistics

Example 4.3.4 A string of 10 light bulbs is connected in series, which means that the entire string will not light up if any one of the light bulbs fails. Assume that the lifetimes of the bulbs, $\tau_1, \dots, \tau_{10}$, are independent random variables that are exponentially distributed with mean 2. Find the distribution of the life length of this string of light bulbs.

4.3 Order statistics

Example 4.3.4 A string of 10 light bulbs is connected in series, which means that the entire string will not light up if any one of the light bulbs fails. Assume that the lifetimes of the bulbs, $\tau_1, \dots, \tau_{10}$, are independent random variables that are exponentially distributed with mean 2. Find the distribution of the life length of this string of light bulbs.

Solution

Note that the pdf of τ_i is $f(t) = 2e^{-2t}$, $0 < t < \infty$, and the cumulative distribution of τ_i is $F_{\tau_i}(t) = 1 - e^{-2t}$. Let T represent the lifetime of this string of light bulbs. Then, $T = \min(\tau_1, \dots, \tau_{10})$.

Thus, $F_T(t) = 1 - [1 - F_{\tau_i}(t)]^{10}$.

Hence, the density of T is obtained by differentiating $F_T(t)$ with respect to t , that is, $f_T(t) = 10f_{\tau_i}(t) [1 - F_{\tau_i}(t)]^9$

$$= \begin{cases} 2(10)e^{-2t}(e^{-2t})^9 = 20e^{-20t}, & 0 < t < \infty \\ 0, & \text{otherwise} \end{cases}$$

4.3 Order statistics

Theorem 4.3.3 Let $X_1, \dots, X_n$ be a random sample with continuous pdf $f(x)$ and a distribution function $F(x)$. Let $Y_1, \dots, Y_n$ be the order statistics. Then for any $1 \leq i < k \leq n$ and $-\infty < x \leq y < \infty$, the joint pdf of Y_i and Y_k is given by:

$$f_{Y_i, Y_k}(x, y) = \frac{n!}{(i-1)! (k-i-1)! (n-k)!} \times \\ [F(x)]^{i-1} [F(y) - F(x)]^{k-i-1} [1 - F(y)]^{n-k} f(x) f(y)$$

4.3 Order statistics

Example 4.3.5 Let $X_1, \dots, X_n$ be a random sample from $U[0, 1]$. Find the joint pdf of Y_2 and Y_5 .

4.3 Order statistics

Example 4.3.5 Let $X_1, \dots, X_n$ be a random sample from $U[0, 1]$. Find the joint pdf of Y_2 and Y_5 .

Solution

Taking $i = 2$ and $k = 5$ in [Theorem 4.3.3](#), we get the joint probability density function (pdf) of Y_2 and Y_5 as:

$$\begin{aligned} f_{Y_2, Y_5}(x, y) &= \frac{n!}{(2-1)!(5-2-1)!(n-5)!} [F(x)]^{2-1} \\ &\quad [F(y) - F(x)]^{5-2-1} \times [1 - F(y)]^{n-5} f(x) f(y) \\ &= \begin{cases} \frac{n!}{2(n-5)!} x(y-x)^2 (1-y)^{n-5}, & 0 < x \leq y < 1 \\ 0, & \text{otherwise} \end{cases} \end{aligned}$$

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4.4 The normal approximation to the binomial distribution

A binomial random variable Y , with parameters n and $p = P(\text{success})$, can be viewed as the number of successes in n trials and can be written as:

$$Y = \sum_{i=1}^n X_i$$

where

$$X_i = \begin{cases} 1, & \text{with probability } p \\ 0, & \text{with probability } 1 - p \end{cases}$$

4.4 The normal approximation to the binomial distribution

- The fraction of successes in n trials is:

$$\frac{Y}{n} = \frac{1}{n} \sum_{i=1}^n X_i = \bar{X}$$

hence, $\frac{Y}{n}$ is sample mean. Since $E(X_i) = p$ and $Var(X_i) = p(1-p)$, we have:

$$E\left(\frac{Y}{n}\right) = E\left(\frac{1}{n} \sum_{i=1}^n X_i\right) = \frac{1}{n} np = p$$

and

$$Var\left(\frac{Y}{n}\right) = \frac{1}{n^2} \sum_{i=1}^n Var(X_i) = \frac{p(1-p)}{n}$$

- By the central limit theorem, $Y = n\bar{X}$ has an approximate normal distribution with mean $\mu = np$ and variance $\sigma^2 = np(1-p)$

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4.5 Chapter summary

- Statistics and the sampling distributions
- Sampling distributions associated with normal populations
 - chi-square distributions
 - T distributions
 - F distributions
- Order statistics
- Large sample approximations such as the normal approximation to the binomial distribution

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4.6 Computer examples

Examples using R (or others)

- [Example 4.6.1](#) Generating normal random samples:

Create three samples of size 30 from standard normal distribution and draw histograms for each sample.

- [Example 4.6.2](#) Generating a normal random sample:

Generate 50,000 observations from a normal distribution with mean 30 and standard deviation 8. Obtain summary statistics for these data and draw a graph

- [Example 4.6.3](#) Generating a random exponential sample

From an exponential distribution, draw 10,000 samples, each sample of size 15. Compute the mean of each sample and draw a chart for the means. This will be an approximate sampling distribution of $\bar{X}$ for a fixed sample of size 15

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Projects for chapter 4

- 4A A method to obtain random samples from different distributions
 - Simulate an observation from the exponential distribution
- 4B Simulation experiments
 - Use simulation experiments to obtain information about a statistic's sampling distribution
- 4C A test for normality
 - the normal-score plot

Exercises and Projects

- Exercises

- Exercises 4.1.3, 4.1.9
- Exercises 4.2.5, 4.2.6, 4.2.15, 4.2.23, 4.26
- Exercises 4.3.1, 4.3.4, 4.3.9

- Projects

- (1) Generating normal random samples (suppose μ, σ^2 fixed)
calculate the median, sample mean and variance with different sample sizes, for example, $n = 100, 500, \text{ or } 1000$ and compare with the mean and variance of population. Draw a table or figure to show them, if necessary.
- (2) Drawing 10,000 samples, each sample of size 15 (suppose μ, σ^2 fixed)
Compute the mean of each sample and draw a histogram for the means. This will be an approximate sampling distribution of $\bar{X}$ for a fixed sample of size 15, and compare with the pdf figure of $\bar{X} \sim N(\mu, \sigma^2/15)$.